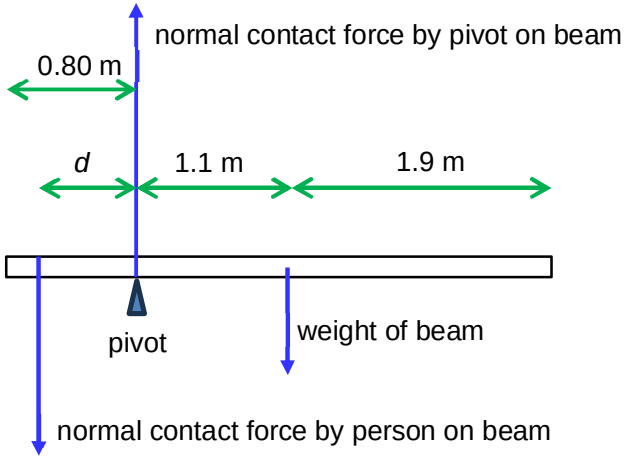


3(a)	The moment of a force about a point is the product of the force and the perpendicular distance from the point to the line of action of the force.
3(b)(i)	 The diagram shows a horizontal beam with a pivot point. A vertical blue arrow labeled "normal contact force by pivot on beam" points upwards from the pivot. A vertical blue arrow labeled "normal contact force by person on beam" points downwards from the left end of the beam. A vertical blue arrow labeled "weight of beam" points downwards from a point 1.1 m to the right of the pivot. Horizontal green double-headed arrows indicate distances: 0.80 m from the left end to the pivot, d from the pivot to the weight, 1.1 m from the pivot to the weight, and 1.9 m from the weight to the right end of the beam.

3(b)(ii)	Taking moment about the pivot, at rotational equilibrium, $(720)(d) = (42 \times 9.81)(1.1)$ $d = 0.63 \text{ m}$
3(b)(iii)	At translational equilibrium, upward force applied by pivot on beam $= 720 + (42)(9.81)$ $= 1132 \text{ N}$ $\approx 1100 \text{ N}$
3(c)(i)	The upward force decreases. Taking moments about the support that is added, the anticlockwise moment due to the person decreases. The total anticlockwise moment decreases. For rotational equilibrium, the clockwise moment due to the pivot decreases. Since the distance 3.0 m is constant, the upward force by pivot on beam decreases.
3(c)(ii)	The upward force decreases. Taking moments about the support that is added, the anticlockwise moment due to the person decreases. The total anticlockwise moment decreases. For rotational equilibrium, the clockwise moment due to the pivot decreases. Since the distance 3.0 m is constant, the upward force by pivot on beam decreases.
4(a)(i)	Frictional force of the road on the bike